

CSCC63 – WEEK 4 TUTORIALS

There are several questions here, you aren't expected to finish all of them today. But they'll be good practice for you.

1. Suppose L is a recognizable language. Show that $L \leq_m A_{\text{TM}}$.
2. Let's practice some mapping reductions and language recognition.

For each of the following languages:

- Write out the language definition using only quantifiers and decidable predicates.

$$\text{E.g., } E_{\text{TM}} = \{ \langle M \rangle \mid \forall w \in \Sigma^*, k \in \mathbb{N}, \langle M, w, k \rangle \notin \text{ACC} \}.$$

- Determine whether it's decidable, recognizable, co-recognizable, or neither.
- Give a sketch for either a recognizer or a mapping reduction that would prove it.

Just for today you can skip the iff section of the proofs – we just want to get used to the process of making the reduction. But it will be a good idea to keep it in mind as you do the reduction, since it'll help guide the process. You should also be prepared to supply the full proof if needed.

- **Thinking ahead:** For which $i \neq j$ do you think you can find a mapping reduction $X_i \leq_m X_j$? We just want the guesses, we'll leave the proofs for later.

- (a) $X_1 = \{ \langle M \rangle \mid \text{There is some } w \in \{0, 1\}^* \text{ such that } M \text{ accepts both } w \text{ and } ww \}$.
- (b) $X_2 = \{ \langle M \rangle \mid \text{There is some } w \in \{0, 1\}^* \text{ such that } M \text{ accepts } w \text{ but loops on } ww \}$.
- (c) $X_3 = \{ \langle M \rangle \mid \text{Whenever } M \text{ accepts } w \in \{0, 1\}^*, M \text{ loops on } ww \}$.
- (d) $X_4 = \{ \langle M \rangle \mid \text{Whenever } M \text{ loops on } w \in \{0, 1\}^*, M \text{ also loops on } ww \}$.

3. For extra practice:

An undirected graph $G = (V, E)$ is 3-colourable iff there is some way to colour each of its nodes red, green, or blue in such a way that no graph edge is monochromatic: i.e., such that there is no red-red edge, no blue-blue edge, and no green-green edge in the graph.

Repeat the PCP question from tutorial 2, but this time set up your PCP instance so it is a yes-instance iff G is an undirected 3-colourable graph.

Note: this is a bit tricky, so I don't intend to use it in the tests. But it's a good exercise, so try it out.